Microeconomic Theory: TA Session

Pinda Wang

Johns Hopkins University

April 3, 2026

Cartels

When firms collude and reduce output in order to raise price (and increase their profits), we say the industry is organized as a **cartel**.

Continuing with the example from last week, suppose there are 2 identical firms in the market. Each firm incurs constant marginal cost of $MC = 2$ and no fixed costs. The market demand curve is $P = 10 - Q$.

1. Suppose the firms engage in Cournot competition. Calculate the equilibrium outputs, price and profits.

   Answer:

   Let q_1, q_2 be the quantities of the two firms, so $Q = q_1 + q_2$.

   The demand curve that firm 1 faces is $P = (10 - q_2) - q_1$. Then, firm 1's revenue $= Pq_1 = (10 - q_2)q_1 - q_1^2$, $MR_1 = (10 - q_2) - 2q_1$.

   Profit maximization means $MR_1 = MC \;\Rightarrow\; (10 - q_2) - 2q_1 = 2$.

   The same with firm 2: $MR_2 = MC \;\Rightarrow\; (10 - q_1) - 2q_2 = 2$.

   Solving the above two equations, we get $q_1 = q_2 = \frac{8}{3}$, $P = 10 - q_1 - q_2 = \frac{14}{3}$. The profit of each firm is $(P - 2)q_1 = \frac{64}{9}$.

Cartels

Suppose there are 2 identical firms in the market. Each firm incurs constant marginal cost of $MC = 2$ and no fixed costs. The market demand curve is $P = 10 - Q$.

2. If the firms can collude and agree to split profits equally, what are the outputs, price and profits?

   Answer: When they collude, they'll behave like a single monopolist.

   $MR = 10 - 2Q = 2 = MC \;\; \Rightarrow \;\; Q = 4, \;\; P = 6.$

   Since they split equally, we have $q_1 = q_2 = 2$. The profit of each firm is $\pi_1 = \pi_2 = Pq_1 - 2q_1 = 8$.

3. Is this an equilibrium? Explain.

   Answer: No. Consider the profit-maximization problem of firm 1 given that firm 2 produces $q_2 = 2$:

   $P = 10 - q_1 - 2 = 8 - q_1, \;\; MR_1 = 8 - 2q_1 = 2 = MC, \;\Rightarrow\;\; q_1 = 3.$

   The new price $P = 5, \;\; \pi_1 = 9$. So, either firm has an incentive to unilaterally increase quantity to 3, which increases its profit from 8 to 9. This is why cartels are often unstable.

Cartels with infinite periods

Is there a way to "enforce" collusion between cartel members?

- Yes, if the game is played for infinite periods.

Firms can employ a **"grim trigger"** strategy:

- Plan to cooperate and produce the monopoly quantity forever
- However, if the other side deviates once, then I will punish the other side by producing the Cournot quantity forever

Cartel with infinite periods

Suppose there are 2 identical firms in the market. Each firm incurs constant marginal cost of $MC = 2$ and no fixed costs. Every year, the market demand curve is $P = 10 - Q$, and the firms choose their quantity. The "discount factor" is $\beta = 0.9$: next year's \$1 is worth \$0.9 today, and 2028's \$1 is worth \$0.81...

1. What is the present value of firm 1's profit if the firm honors the cartel agreement forever?

   Answer: From question (2) in the previous question, monopoly profit of each firm in each period is \$8. So, the present value of firm 1's profit is $8 + 0.9 \times 8 + 0.9^2 \times 8 + 0.9^3 \times 8 ... = \frac{8}{1-0.9} = 80$.

2. Suppose the grim trigger strategy is used. What is the present value of firm 1's profit if firm 1 decides to deviate in the first period?

   Answer: Firm 1 chooses to produce the profit-maximizing quantity as in question (3) of the previous question. First-period profit is 9.

   Afterwards, the grim trigger is activated, and the two firms produce Cournot quantity forever. The profit of subsequent periods is $\frac{64}{9}$.

   Present value of profit is $9 + 0.9 \times \frac{64}{9} + 0.9^2 \times \frac{64}{9} + ...$
   $= 9 + \frac{0.9 \times 64/9}{1-0.9} = 73$.

Cartel with infinite periods

Suppose there are 2 identical firms in the market. Each firm incurs constant marginal cost of $MC = 2$ and no fixed costs. Every year, the market demand curve is $P = 10 - Q$, and the firms choose their quantity. The "discount factor" is $\beta = 0.9$: next year's \$1 is worth \$0.9 today, and 2028's \$1 is worth \$0.81...

3. Can the two firms enforce the cartel agreement?

   Answer: Yes, because the present value of profits from cooperating (\$80) is greater than the profits from undermining the cartel (\$73).

4. With what values of the discount factor β can the two firms enforce the cartel agreement? Explain intuitively.

   Answer: The present value of a firm's profit from sticking to the cartel is $8 + \beta \times 8 + \beta^2 \times 8 + ... = \frac{8}{1-\beta}$.

   The present value of a firm's profit from undermining the cartel in the first period is $9 + \beta \times \frac{64}{9} + \beta^2 \times \frac{64}{9} + ... = 9 + \frac{64\beta}{9(1-\beta)}$.

   The cartel can be enforced if $\frac{8}{1-\beta} > 9 + \frac{64\beta}{9(1-\beta)} \Rightarrow \beta > \frac{9}{17}$.

   Intuitively, firms have to be "patient" enough to care about profits far in the future.